

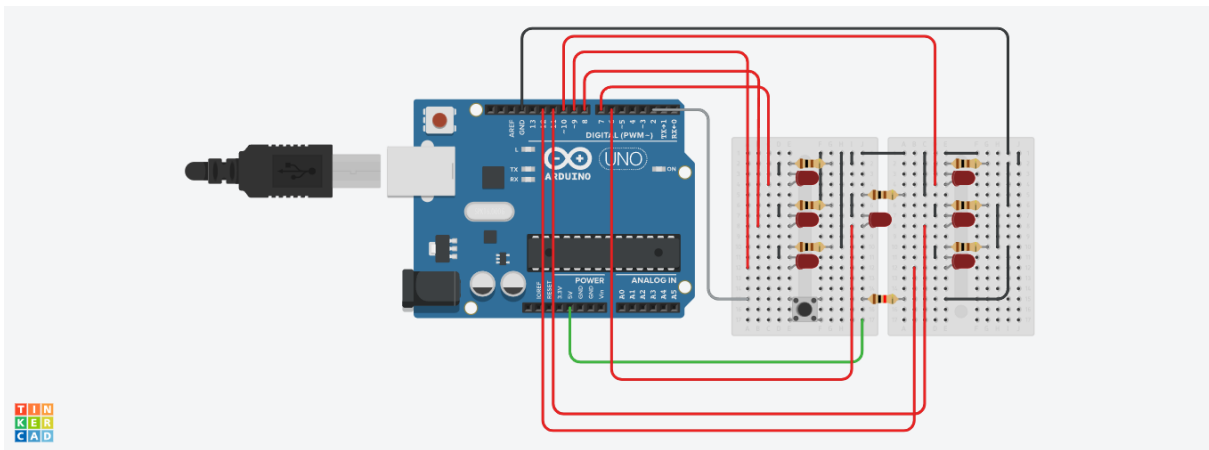
Sistemas Embarcados e IOT

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Turma: 1ºc2

Atividade 3 – Dado eletrônico

Imagem do arduino:



Código:

```
// INIT IDENT
int pinoLed1 = 7;
int pinoLed2 = 8;
int pinoLed3 = 9;
int pinoLed4 = 10;
int pinoLed5 = 11;
```

```
int pinoLed6 = 12;
```

```
int pinoLed7 = 6;
```

```
int pinoBotao = 2;
```

```
void setup() {
```

```
    pinMode(pinoLed1,OUTPUT);
```

```
    pinMode(pinoLed2,OUTPUT);
```

```
    pinMode(pinoLed3,OUTPUT);
```

```
    pinMode(pinoLed4,OUTPUT);
```

```
    pinMode(pinoLed5,OUTPUT);
```

```
    pinMode(pinoLed6,OUTPUT);
```

```
    pinMode(pinoLed7,OUTPUT);
```

```
    pinMode(pinoBotao,INPUT_PULLUP);
```

```
    randomSeed(analogRead(0));
```

```
}
```

```
// FIM IDENT
```

```
// SORTEIO
```

```
void loop() {
```

```
    if(digitalRead(pinoBotao) == HIGH) {
```

```
        sorteia();
```

```
    }
```

```
}
```

```
void sorteia() {
```

```
    int numeroSorteado = random(1,8);
```

```
    digitalWrite(pinoLed1, LOW);
```

```
    digitalWrite(pinoLed2, LOW);
```

```
digitalWrite(pinoLed3, LOW);  
digitalWrite(pinoLed4, LOW);  
digitalWrite(pinoLed5, LOW);  
digitalWrite(pinoLed6, LOW);  
digitalWrite(pinoLed7, LOW);
```

```
if(numeroSorteado==1) {  
    digitalWrite(pinoLed7, HIGH); //SE SORT=1  
}
```

```
if(numeroSorteado==2) {  
    digitalWrite(pinoLed1, HIGH); //SE SORT=2  
    digitalWrite(pinoLed6, HIGH);  
}
```

```
if(numeroSorteado==3) {  
    digitalWrite(pinoLed3, HIGH);  
    digitalWrite(pinoLed7, HIGH); //SE SORT=3  
    digitalWrite(pinoLed4, HIGH);  
}
```

```
if(numeroSorteado==4) {  
    digitalWrite(pinoLed1, HIGH);  
    digitalWrite(pinoLed3, HIGH); //SE SORT=4  
    digitalWrite(pinoLed4, HIGH);  
    digitalWrite(pinoLed6, HIGH);  
}
```

```
if(numeroSorteado==5) {
```

```
digitalWrite(pinoLed1, HIGH);  
digitalWrite(pinoLed3, HIGH);  
digitalWrite(pinoLed6, HIGH); //SE SORT=5  
digitalWrite(pinoLed7, HIGH);  
digitalWrite(pinoLed4, HIGH);  
}
```

```
if(numeroSorteado==6) {  
    digitalWrite(pinoLed1, HIGH);  
    digitalWrite(pinoLed2, HIGH);  
    digitalWrite(pinoLed3, HIGH); //SE SORT=6  
    digitalWrite(pinoLed5, HIGH);  
    digitalWrite(pinoLed6, HIGH);  
    digitalWrite(pinoLed4, HIGH);  
}
```

```
delay(2000);
```

```
// RESET
```

```
digitalWrite(pinoLed1, LOW);  
digitalWrite(pinoLed2, LOW);  
digitalWrite(pinoLed3, LOW);  
digitalWrite(pinoLed4, LOW);  
digitalWrite(pinoLed5, LOW);  
digitalWrite(pinoLed6, LOW);  
digitalWrite(pinoLed7, LOW);  
}
```